SABIL

Solar Access Builds Independent Lives

A home solar kit whose controller makes the battery last, sold per kWh to families in Maiduguri who lose the grid for months at a time.

Brief
13. Power for a city the grid can't protect

Track and setting
4. Clean Energy Solutions; A. Residential

Team
Youcef Toukal (batteries) and Younes Toukal (AI, software)

Build files
github.com/YounesToukal/SABIL

Community and target user

Five-person households in Maiduguri, Borno State. The city depends on one 330 kV line through insurgent-held land; in 2021 it spent about ten months without grid power, and attacks on Nigerian towers doubled from 2024 to 2025 (Brief 13). The family pays per kWh; the operator owns the kit.

The problem

When the line is cut, families fall back on generators (petrol, $1.07/L in Borno) or on solar kits. A standard kit charges its lithium battery to 100% and leaves it full all afternoon at 35 to 44 °C. That is how lithium batteries wear fastest, so the kit fades and the essentials (lights, phones, fan, fridge) start going dark from about the fourth year.

Proposed solution

A 400 Wp, 1.92 kWh LFP kit with a controller that knows how batteries age. Each morning it forecasts tomorrow's sun; on hot days with a sunny tomorrow it charges only what tonight needs, keeps a reserve for essentials, serves them first and moves flexible loads to sunny hours. Top-ups are paid by phone, and the operator guarantees battery health, so it has every reason to make the battery last.

Proof of concept built in 48 hours

A 10-year simulation on NASA POWER hourly weather for Maiduguri (2014 to 2023), with a battery-ageing model that reproduces the TU Munich reference code to 4.5×10^{-14} and a second NREL model as a cross-check, a forecasting lab (12 models), a cost model, and a working controller interface that replays real weather hour by hour. Everything is reproducible from the repository.

Brief metric moved: same kit, same house, same weather, only the controller differs

110 → 13 h
hours without essential power over 10 years (standard → SABIL)

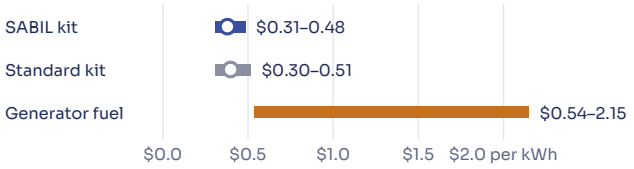
6.2 → 7.5 y
battery life to 80%; SABIL better in 96 of 96 Monte Carlo cases (gain median 1.1 y, 90% range 0.3 to 3.4)

212–849 L
fuel avoided per home per year (generator at 0.5 to 2 L/kWh)

542
battery modules (9.8 t) avoided per 1,000 homes over 20 years

Estimated cost and who pays

Kit $638 (the controller adds $12, repaid in 1.3 years by slower battery wear). Break-even tariff $0.38/kWh (₦505), covering the kit over 20 years at 12% cost of capital, running costs, fees and 10% unpaid bills. A family pays $4.06 a month for lights, phones and fans, $13.40 with a fridge (minimum wage: $53), nothing up front. A DARES grant (~$84 a kit) lowers the tariff to $0.35.



Ranges: battery $120–268/kWh, capital 8–18%, unpaid bills 5–20%, use ±20%, two ageing models; petrol at the Borno pump, $1.07/L (₦1,430); ₦1,332 per US dollar. Dot: central case.

6 to 18 month deployment path

- **Months 0–3:** partner with a DARES-registered solar company in Borno; source VeraSol-certified parts; flash the controller onto 50 kits.
- **Months 3–9:** 50-home pilot in Maiduguri; log charge, temperature and essential hours every hour; local agents trained to install and service.
- **Months 9–18:** refit the ageing model on pilot data; scale to 1,000 homes with the partner's pay-per-use platform and DARES grants.

How it scales

The controller is firmware plus $12 of sensors, so existing kit makers can adopt it without new hardware lines. Next: Borno, Yobe and Adamawa, where lines are attacked, then the hot off-grid Sahel. Unit cost falls with battery prices (LFP packs −8% in 2025, BNEF) and with volume. A second city only needs its weather data and local prices.

Biggest risk and how we test it

The ageing model is fitted on lab cells, not on our packs. The pilot logs each battery's charge and temperature hourly, so we can refit it (semi-empirical first, then an electrochemical P2D model) on field data. We also tried tuning the rules by Bayesian optimization on unseen 2019–2023 weather: longer battery life came only with more lost essential hours, so we kept the reliable rules.

Prior work disclosure. Before 30 October 2026 we had only our training: Youcef's MSc courses in photovoltaics and battery modelling, Younes's machine-learning background. All models, data processing, results, figures and the interface were built during the sprint. Open data and code used: NASA POWER, ERA5, NOAA climate indices, SimSES coefficients (TU Munich), NREL BLAST-Lite, Optuna. Prices: NNPC (Oct 2026), NBS (Mar 2026), CBN rate ₦1,332/USD, PVPRO Lagos (Sep 2026), BNEF (Dec 2025), World Bank DARES appraisal (2023), A2EI generator study (2022).